SAP ABAP CAPSTONE PROJECT

PROJECT TITLE:

CUSTOMER ORDER & PAYMENT TRACKING SYSTEM

USING SAP ABAP (O2C)

Submitted by:

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Batch: 2023–2027

1. INTRODUCTION

The Order-to-Cash (O2C) process is a key part of the SAP Sales and Distribution (SD) module, covering everything from when a customer places an order to when the payment is received. It plays an important role in managing business transactions and ensuring smooth financial operations.

In this project, the O2C process is recreated using SAP ABAP concepts in a simulated setup. Since access to a real SAP system was not available, internal tables have been used to represent standard SAP tables and transactions.

The project shows how different business activities—such as sales orders, billing, and payments—are connected and processed through a custom ABAP report. It also includes the simulation of standard SAP tables like KNA1, VBAK, VBRK, and BSID using internal data structures.

2. PROBLEM STATEMENT

In practical business scenarios, tracking customer payments and identifying delays can be challenging, especially when dealing with large volumes of transactional data. Standard SAP reports often present detailed information, but they may not always be convenient for quick analysis or decision-making.

Therefore there is a need for a simplified reporting system that can:

- Track customer transactions from start to end
- Calculate payment delays efficiently
- Identify customers who may pose a financial risk
- Present information in a clear and well-structured format
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3. OBJECTIVE

The objective of this project is to:

- Simulate the complete Order-to-Cash (O2C) process in a simple and understandable way
- Develop a custom ABAP report to display transaction details
- Analyze customer payment behavior based on transaction data

- Calculate payment delays and classify customers based on risk levels
- Replicate SAP-like system functionality using internal tables instead of a real database

4. SOLUTION OVERVIEW

This project provides a simulated SAP O2C system using ABAP programming. It integrates customer, order, invoice, and payment data into a unified reporting system.

Key features include:

- End-to-end O2C process simulation
- Automated delay calculation
- Risk-based customer classification
- Interactive filtering system
- Structured report output similar to SAP ALV

5. SCOPE OF PROJECT

This project covers:

- Customer Master Data
- Sales Order Processing
- Invoice Generation
- Payment Recording
- Report Generation

The project is implemented using internal tables and modular ABAP programming.

6. O2C PROCESS FLOW

The following SAP transactions are simulated in this project:

Steps	Process	T-Code
1	Customer Creation	XD01
2	Sales Order Creation	VA01
3	Delivery Processing	VL01N

4	Billing (Invoice)	VF01
5	Payment Posting	F-28
6	Customer Line Items	FBL5N

The process flow:

Customer → Sales Order → Delivery → Invoice → Payment → Report

7. SYSTEM DESIGN

7.1 Data Simulation

Instead of real SAP tables, the following internal tables are used:

- Customer Table (simulates KNA1)
- Sales Order Table (simulates VBAK)
- Invoice Table (simulates VBRK)
- Payment Table (simulates BSID)

7.2 ABAP Program Structure

The program is divided into multiple sections:

- Type Definitions
- Internal Tables
- Data Loading
- Business Logic
- Filtering
- Output Display

8. FUNCTIONALITY

8.1 Data Linking

The program connects:

- Sales Orders → Invoices
- Invoices → Payments

8.2 Delay Calculation

Payment delay is calculated as:

Delay = Payment Date – Invoice Date

8.3 Risk Classification

Delay Days	Status
0-5	On Time
6-15	Late
>15	High Risk

8.4 User Filters

The report allows filtering by:

- Customer ID
- Payment Status
- Delayed Payments

8.5 Output Display

The report is displayed in a structured format similar to SAP ALV reports, showing:

- Customer details
- Order information
- Invoice data
- Payment status
- Delay days

9. TECHNOLOGIES USED

- SAP ABAP – Backend logic and report generation
- HTML/CSS – Frontend simulation interface
- VS Code – Development environment
- GitHub – Version control and project hosting

10. UNIQUE FEATURES

- Simulation of SAP system without real access
- End-to-end O2C process implementation
- Automated delay calculation
- Risk-based classification
- User-friendly report output
- Modular and clean ABAP code

11. FUTURE ENHANCEMENTS

- Integration with real SAP database
- ALV Grid with interactive features
- Graphical dashboard (charts)
- Email alerts for delayed payments
- Integration with SAP Fiori UI

12. CONCLUSION

This project presents a practical implementation of the Order-to-Cash process using SAP ABAP in a simulated environment. It helps in understanding how different stages of business transactions are connected and how both functional and technical aspects work together.

Through this project, it becomes easier to analyze customer payments, identify delays, and recognize customers who may pose a risk. These insights are useful for improving decision-making in business scenarios.

Overall, the project reflects a good understanding of SAP concepts and can be further enhanced to work in real-time systems with additional features and integrations.

13. SCREENSHOTS

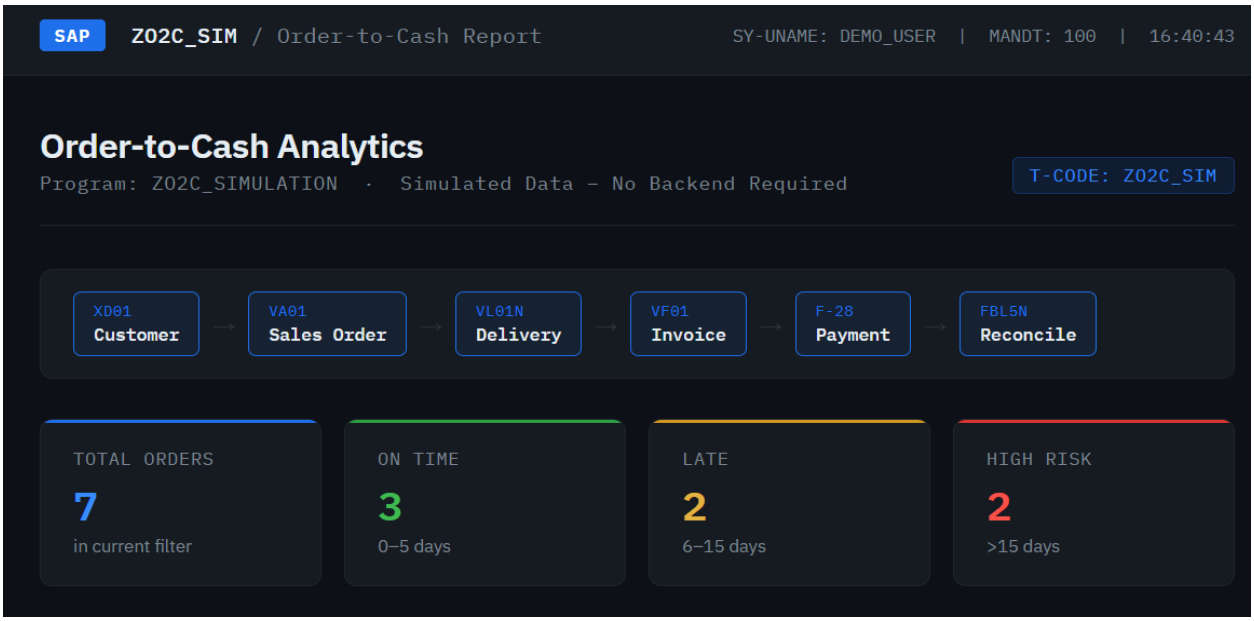

Figure 1: O2C Dashboard showing process flow

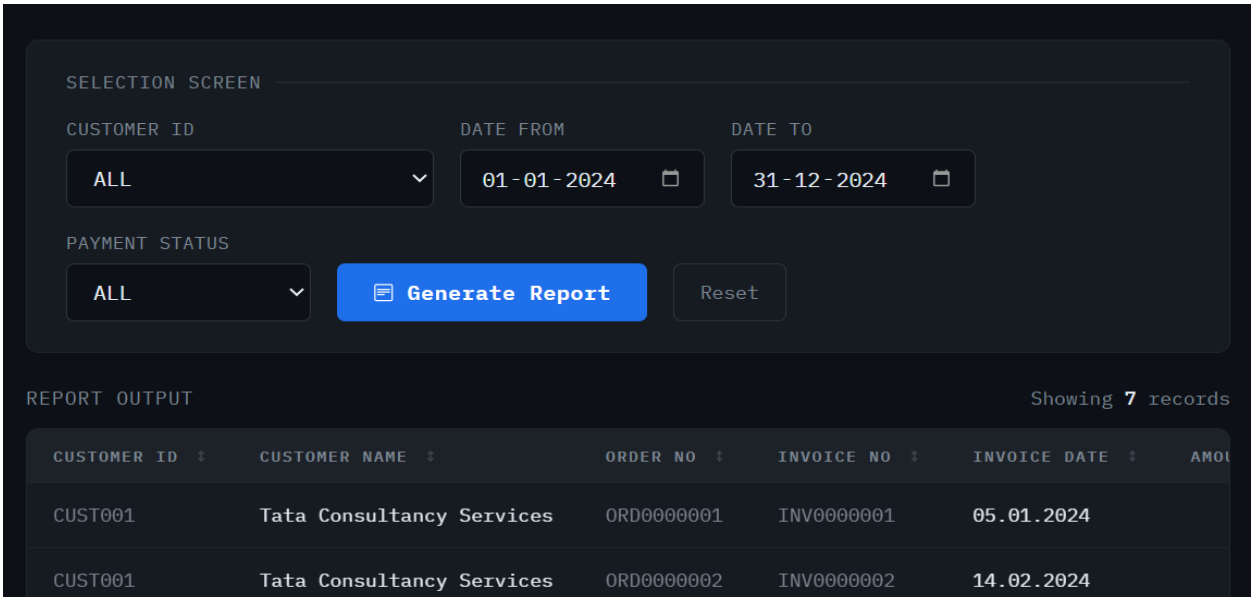

Figure 2: User input screen for filtering report data

CUSTOMER ID	CUSTOMER NAME	ORDER NO	INVOICE NO	INVOICE DATE	AMOU
CUST001	Tata Consultancy Services	ORD0000001	INV0000001	05.01.2024	
CUST001	Tata Consultancy Services	ORD0000002	INV0000002	14.02.2024	
CUST002	Infosys Limited	ORD0000003	INV0000003	18.01.2024	
CUST003	Wipro Technologies	ORD0000004	INV0000004	24.01.2024	
CUST004	HCL Technologies	ORD0000005	INV0000005	05.02.2024	
CUST004	HCL Technologies	ORD0000006	INV0000006	18.02.2024	
CUST005	Tech Mahindra Ltd	ORD0000007	INV0000007	10.02.2024	

ZO2C_SIMULATION · Simulated Data Only – No SAP DB Tables Used INR · MANDT 100 · ERP 6.0 EHP8

Figure 3: Generated O2C report displaying customer transactions and delay analysis